

Biomaterial Design Inspired by Regenerative Research Organisms

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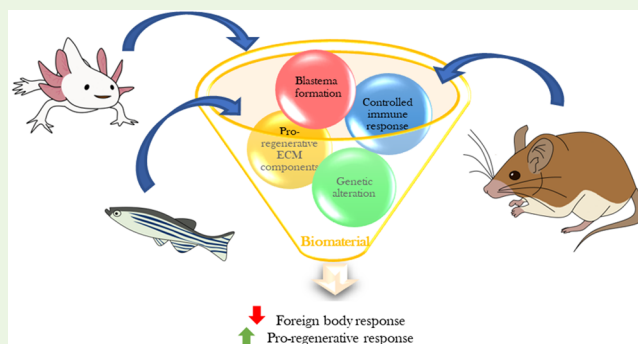
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ABSTRACT: The efficacy of implanted biomaterials is largely dependent on the response of the host's immune and stromal cells. Severe foreign body response (FBR) can impede the integration of the implant into the host tissue and compromise the intended mechanical and biochemical function. Many features of FBR, including late-stage fibrotic encapsulation of implants, parallel the formation of fibrotic scar tissue after tissue injury. Regenerative organisms like zebrafish and salamanders can avoid fibrosis after injury entirely, but FBR in these research organisms is rarely investigated because their immune competence is much lower than humans. The recent characterization of a regenerative mammal, the spiny mouse (*Acomys*), has inspired us to take a closer look at cellular regulation in regenerative organisms across the animal kingdom for insights into avoiding FBR in humans. Here, we highlight how major features of regeneration, such as blastema formation, macrophage polarization, and matrix composition, can be modulated across a range of regenerative research organisms to elucidate common features that may be harnessed to minimize FBR. Leveraging a deeper understanding of regenerative biology for biomaterial design may help to reduce FBR and improve device integration and performance.

KEYWORDS: biomaterials, foreign body response, fibrosis, regeneration



1. INTRODUCTION

Tissue repair in nonregenerative organisms such as humans typically results in excessive deposition of collagenous matrix by fibroblasts, leading to the formation of scar tissue. This scar tissue exhibits mechanical properties that differ from those of the surrounding tissue, which can cause impairment of normal function. In contrast, regenerative organisms replace damaged tissue with native architecture that maintains the functionality of the original tissue. Regeneration among living beings exists on a spectrum in which few organisms are bestowed with the gift of self-renewal. This spectrum exists due to the loss of regenerative abilities that occurred during evolution,² although the mechanisms behind this evolution have been hotly debated since the 1990s.³ Only lower organisms such as hydra and planaria are capable of whole-body regeneration. Among vertebrates, regeneration is less extensive, with certain lower vertebrates including zebrafish and salamanders possessing the ability to regenerate lost appendages and other organs. Regeneration in response to injury is not common among mammals, but there are some exceptions, including ear pinna regeneration in rabbits,^{4–6} some tissues in the MRL mouse strain,^{7,8} digit tips in rodents,⁹ and several tissues in *Acomys*.¹ Regeneration following injury is even more limited in humans.

Biomaterials can be leveraged to address this disparity in regenerative capabilities in humans by replacing or improving the function of damaged tissues.^{10,11} For example, bionic limbs

can partially restore lost function through the replacement of the damaged limb.¹² Similarly, pacemakers are used to control irregular heart rhythms and improve the function of arrhythmic hearts.¹³ The biggest challenge facing the long-term success of these medical implants is a foreign body response (FBR) in the recipient. FBR is a local and systemic response initiated by the host immune system against the foreign material and can be so severe that it compromises the strength, efficacy, and functionality of the implant which can potentially cause complete failure.

Fibrotic encapsulation observed in the late stage of FBR is similar to the excessive matrix deposition commonly observed after injury in nonregenerative organisms. Therefore, our knowledge of regenerative organisms that avoid this excessive matrix deposition may be used to inform the design of biomaterials to mitigate FBR. We have recently supported this potential link between regeneration and reduced FBR in *Acomys* through *in vivo* experimentation. When challenged with a subcutaneous polydimethylsiloxane implant, *Acomys* demon-

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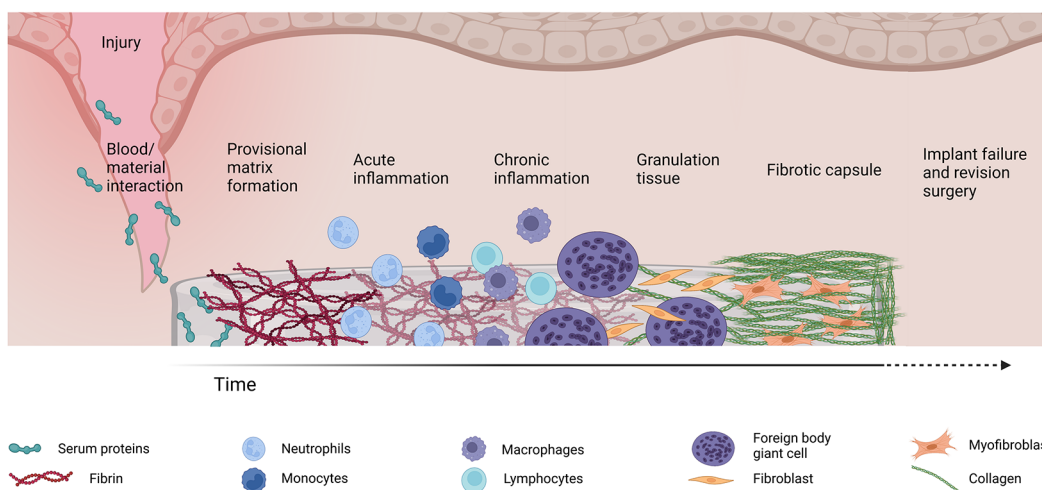


Figure 1. Foreign body reaction begins upon injury to the host tissue. As the biomaterial is implanted, a blood/material interaction occurs during which serum proteins including complement factors adsorb to the surface of the implant and recruit immune cells. A provisional matrix dominated by fibrin also forms due to the blood/material interaction. The acute inflammation phase of FBR is characterized by neutrophil recruitment, monocyte recruitment, and differentiation of monocytes into macrophages. If inflammation persists, chronic inflammation occurs in which lymphocytes are recruited and macrophages turn into foreign body giant cells. Fibroblasts are recruited and begin to generate granulation tissue. Depending on macrophage phenotype, fibroblasts can be activated into myofibroblasts and deposit collagen-rich, dense ECM that forms a fibrotic capsule that may ultimately result in implant failure and a revision surgery that restarts the process. Created with BioRender.com.

strated reduced monocyte infiltration, fibroblast activation, and fibrotic encapsulation thickness around the implant when compared to common mice (*Mus musculus*).¹⁴ In this review, we will identify distinctive features reported in *Acomys* and other regenerative organisms that may inform the design of biomaterials tailored to reduce FBR and promote reconstructive healing following implantation.

2. OVERVIEW OF FBR STAGES

FBR consists of a series of events that are triggered by the host in response to an external material. Multiple properties of the biomaterial can prompt this chain reaction including antigenic properties, degradation rate and products, surface chemistry and topography, and mechanical properties. FBR commences with the adsorption of proteins and complement factors from blood plasma, which act as signals to recruit immune cells to clear out the foreign body. This interaction with the blood results in thrombus or blood clot formation on the surface of the material which can be described as a provisional matrix.¹⁵ Generation of acute inflammation by immune cells is followed by chronic inflammation, granulation tissue development, and fibrotic encapsulation (Figure 1).

Upon injury, the first cells to arrive at the damaged site are neutrophils, followed by monocytes. Circulatory monocytes are differentiated into tissue adherent macrophages in the presence of colony-stimulating factors, which have been identified as essential regulators of fibrosis.¹⁶ Macrophages are the key immune cells involved in the inflammatory response; they mediate the degradation of the biomaterial by phagocytosis. The biocompatibility of the biomaterial and other physical-mechanical traits determines the extent of the phagocytosis and the severity of the FBR. An interesting feature of macrophages is their polarization state, which exists within a spectrum between classically activated (M1) and alternatively activated (M2) states and dictates the course of an immune response. This balance between macrophage polarization states is tightly regulated in native tissue. Classically

activated, pro-inflammatory macrophages are typically activated at the onset of injury and promote a phagocytotic environment that is associated with cytokines such as IL-1 β , IL-6, and TNF- α . On the other hand, anti-inflammatory macrophages (M2-like) are mainly involved in the repair and produce a different set of cytokines including IL-4 and IL-10. During the initial response to an implanted material, IFN- γ produced by natural killer cells activates macrophages to produce inflammatory cytokines. This increases the infiltration of TH17 cells, which produce IL-17, causing damage by enhancing the inflammatory response.¹⁷ The multinucleated giant cells formed by the fusion of macrophages, also known as foreign body giant cells (FBGCs), are involved in the later stages of FBR to eliminate the foreign body from the system by degradation. FBGCs may also play a detrimental role by supporting chronic FBR. Following biomaterial implantation, an M2-dominated response has largely been associated with tissue remodeling and the absence of persistent inflammation while an M1-dominated response has been associated with long-term inflammation, the formation of FBGCs, and fibrotic tissue.¹⁸ However, these definitions of pro-inflammatory (M1) and anti-inflammatory (M2) macrophages do not capture the complexity of macrophage phenotype during fibrosis or regeneration, and subsets of M2 macrophages have been linked to fibrotic diseases and fibrous capsule thickness during FBR.^{19–21} Much of the literature, including examples included in this review, identifies the macrophage phenotype using markers like CD206 and CD163 that are expressed by several M2 subtypes.²² It is likely that regulation of FBR is led by subsets of macrophage phenotypes that do not fit neatly into the M1 or M2 classification and that the temporal regulation of macrophage phenotypes is also crucial.²³

Another key cell type that contributes to FBR is fibroblasts, which regulate the deposition of extracellular matrix proteins including collagen. Upon injury, fibroblasts are recruited by cytokines and chemokines released by macrophages. A portion of fibroblasts will differentiate into highly contractile myofibroblasts, which help to close the wound through tension

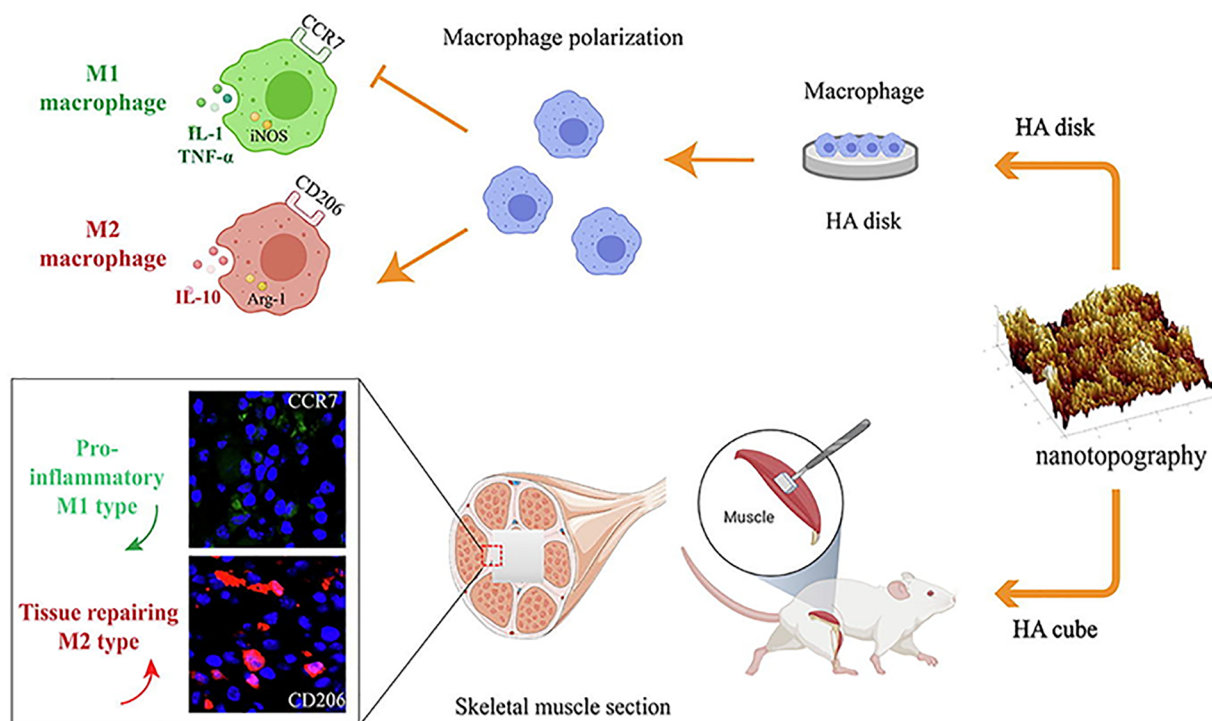


Figure 2. Surface nanotopography can be used to drive the switch of macrophage phenotype to the anti-inflammatory M2 type *in vitro* and *in vivo* to promote tissue regeneration. Hydroxyapatite disks with nanotopography promoted RAW 264.7 macrophage phenotype switching *in vitro* from M1 to M2 when compared to submicrometer topography. In a mouse intramuscular implantation model, nanotopography was associated with an increase in M2/M1 macrophages. Reproduced with permission from ref 39. Copyright 2022 Elsevier.

and deposition of granulation tissue. In response to a biomaterial, myofibroblasts can contribute to excessive deposition of collagen, which creates fibrotic encapsulation around the biomaterial, separating it from the host environment. Damage-associated stress increases the levels of reactive oxygen species and TGF- β , triggering a heightened accumulation of myofibroblasts and leading to the formation of granulation tissue and deposition of excessive collagen, creating a fibrous mesh around the material, and marking the final stage of FBR. In extreme cases, the material is surrounded by fibrotic encapsulation causing an absolute separation from the host. The extent of FBR is vital to determine the fate of the implant, so efforts are being made to improve the designs of biomaterials to reduce the extent of FBR. Below, we discuss such methods based on the information from regenerative animals.

3. SALIENT PROPERTIES FOR BIOMATERIAL DESIGN

Biomaterial systems may be composed of either synthetic or naturally derived materials. Synthetic materials include polymers like polyesters and are highly tailorable and easily accessible but suffer from negative host responses, including inflammatory responses and nonintegration with host tissues due to their foreignness.²⁴ One source of host responses to synthetic materials is degradation products that can result in an inflammatory cascade, as is the case with polyglycolic acid and poly(lactide-co-glycolide).^{25,26} Naturally derived biomaterials can be composed of materials like ECM proteins or peptides, polysaccharides, or decellularized tissues and can readily present complex combinations of protein binding sites and other signals to direct host cell functions, but they also pose the problem of immunogenicity if not properly sourced or

treated.^{24,27,28} Host immune responses to natural biomaterials can stem from the presence of xenogeneic proteins like the Gal epitope, or from residual DNA in the case of decellularized tissues.^{27,28} It is unclear whether FBR is reduced in response to naturally derived biomaterials compared to synthetic biomaterials, and both have advantages and disadvantages.^{27,29} The choice of a synthetic, natural, or combination biomaterial will be influenced by the implant site and desired properties of the materials, but there are many design choices that can be made once a material is chosen that can mediate the host response. Here, we will discuss how host responses to biomaterials can be mediated through modifications to surface topography, surface chemistry, and mechanical properties while paying special attention to cell types involved in regeneration.

3.1. Surface Topography and Surface Chemistry.

Surface topography and chemistry can be modified to alter the cellular processes and phenotypes of the host cells. In general, hydrophilic coatings, increased roughness, and 3D porous structures enhance biocompatibility. Cells tend to attach, proliferate, and remain viable on surfaces that offer a microenvironment mimicking tissue-specific condition, so incorporating cell-adhesion motifs like RGD or coating with a hydrophilic polymer or extracellular matrix components are common approaches to enhance the effectiveness of implants.³⁰ Providing surface cues to modulate the behavior of regeneration-mediating cells like macrophages and stem cells may promote reconstructive healing following implantation of a biomaterial.

The surface topography and chemistry of biomaterials have been shown to modulate macrophage adhesion and polarization^{31–33} through regulation of processes such as the NF- κ B

pathway.³⁴ For example, Barth et al. demonstrated that RAW 246.7 macrophages cultured on roughened titanium disks secreted cytokines more representative of an M2-like macrophage phenotype when compared to macrophages cultured on polished disks.³⁵ Similarly, electrospun scaffolds with increasing fiber diameter and pore size promoted increased M2 marker expression and decreased M1 marker expression in mouse bone marrow-derived macrophages,³⁶ and micropillars with appropriate diameter and spacing modulated the differentiation of human peripheral blood mononuclear cells to an M2 macrophage phenotype.³⁷ More recently, nanostructures have been shown to activate the M2 macrophage phenotype and improve integration *in vivo* when compared to microstructures, as demonstrated by Wang et al. (Figure 2).^{38,39} When hydroxyapatite disks with nano- and microtopography were used in a mouse intramuscular implantation model, nanotopography was associated with an increase in the ratio of M2/M1 macrophages recruited to the biomaterial.³⁹ Since an increased M2/M1 ratio has been linked to both regeneration^{40,41} and constructive remodeling following biomaterial implantation,⁴² modulation of biomaterials to encourage the transition from an M1 to an M2 phenotype may assist with biomaterial integration.

Surface properties can also impact the differentiation and immunomodulatory potential of stem cells.^{43,44} For example, mesenchymal stem cells cultured on fibrous scaffolds produced significantly higher levels of anti-inflammatory cytokines when compared to those cultured on a flat surface.⁴⁵ Topographical cues also impact the expression of the *Zscan4* gene, which is associated with stem cell pluripotency and self-renewal, indicating the relevance of surface–cell contact in directing cellular behavior.⁴⁶ In addition to topographical cues, surface chemistries impact the capacity of stem cells to self renew and have been modulated to maintain stem cell pluripotency during *in vitro* culture.^{47,48} For example, materials generated from monomers with high acrylate content and moderate wettability were demonstrated to be useful as a feeder-free substrate to support pluripotency of embryonic stem cells.⁴⁸ The knowledge gained from these *in vitro* culture methods may inform methods for inducing or maintaining pluripotency *in vivo*, a characteristic important for functional repair within regenerative organisms. Leveraging our knowledge of surface modifications and incorporating our knowledge of regeneration may allow us to fabricate biomaterials with surface properties that guide host cells toward functional repair following biomaterial implantation.

3.2. Mechanical Properties. The mechanical properties of a biomaterial can also play a role in the host reaction upon implantation. Substrate stiffness regulates cell behaviors including adhesion,^{49,50} motility,⁵¹ differentiation of stromal cells,^{52–54} and activation of immune cells.^{55,56} Substrate mechanical properties have long been investigated as a strategy to mitigate FBR, especially for neural electrodes,⁵⁷ with the hopes that softer substrates will reduce macrophage and fibroblast activation. Substrate stiffness influences macrophage polarization,^{31,33} which can result in the failure of surgical implants in cases of mechanical mismatch.⁵⁸ Blakney et al. demonstrated that stiffer hydrogels promoted increased macrophage spreading and activation *in vitro* and a more severe FBR *in vivo*.⁵⁵ Recently, Carnicer-Lombarte et al. demonstrated that FBR can be reduced subcutaneously and in peripheral nerves by matching the mechanical properties of the implant or its coating with that of the host tissue.⁵⁹ Similarly,

stiffer collagen gels have been shown to increase the fusion rate of bone marrow derived monocytes *in vitro* and promote the formation of foreign body giant cells when implanted intramuscularly in rats.⁶⁰ Regulation of macrophage accumulation and foreign body giant cell fusion has recently been linked to the mechanosensory ion channel, TRPV4, supporting the role of substrate mechanics in FBR.⁶¹

The mechanical properties of the biomaterial also have a direct impact on fibroblast activation, which contributes to the deposition of fibrous ECM around biomaterials. Stiffer implanted substrates, in combination with TGF- β signaling, promote fibroblast differentiation into myofibroblasts and increased deposition of a collagenous matrix.⁶² The newly deposited matrix can promote a positive feedback loop where myofibroblast activation is maintained by increased mechanical stiffness, leading to further ECM deposition. This mechanism is a common characteristic of progressive fibrosis.⁶³ Nuskovicova et al. recently demonstrated that coating conventionally stiff silicones with a softer silicone layer reduced collagen deposition and myofibroblast activation around the implant. Local delivery of a small-molecule inhibitor of TGF- β signaling accomplished the same outcome.⁶²

Substrate stiffness regulates the behavior of stem cells as well and has been leveraged in biomaterial scaffolds to maintain potency^{64–66} or direct differentiation of stem cells.^{52,67,68} These biomaterial systems recapitulate properties of the stem cell niche and provide appropriate cues for the lineage of interest. Softer substrates tend to support the self-renewal and pluripotency of embryonic stem cells due to reduced cell spreading and the retention of transcription factors like Oct-4 and Nanog.^{64,66} In addition, soft substrates have been utilized to reprogram mesenchymal stem cells into induced pluripotent stem cells *in vitro*,^{69,70} suggesting a path toward recapitulating blastema-like behaviors to promote functional repair following implantation of a material.

Our knowledge of regeneration-competent organisms and tissues suggests that mechanical stiffness may play a role in regenerative capacity *in vivo* through mechanisms like those described above. For example, the skin of the regenerative *Acomys* has an elastic modulus that is 20 times lower than that of the nonregenerating *Mus musculus*.⁷¹ Similar differences are seen in the stiffness of *Mus* and *Acomys* skeletal muscle.⁷² Notari et al. identified tissue stiffness as an important factor for cardiac regeneration in neonatal mice following apical resection, and pharmacological inhibition of lysyl oxidase significantly decreased cardiac stiffness and rescued the ability of older neonatal mice to mount a regenerative response.⁷³ Decreased tissue stiffness has also been implicated in improved hair regeneration. In a wound-induced hair follicle neogenesis model, softening the wound bed with blebbistatin treatment increased hair neogenesis in mice.⁷⁴ In humans, the soft oral mucosa displays higher regenerative potential than other epithelial wounds, with the ability to execute accelerated, scarless healing.⁷⁵ Similarly, healing of the soft palate is more rapid than healing of the hard palate due to a more robust Wnt and proliferative response, which is thought to be influenced by the different mechanical environments of the two palates.⁷⁶ These examples emphasize the importance of the mechanical environment in functional repair in both regenerative and nonregenerative organisms and the need to design biomaterial systems or coatings that provide appropriate mechanical cues to resident cells.

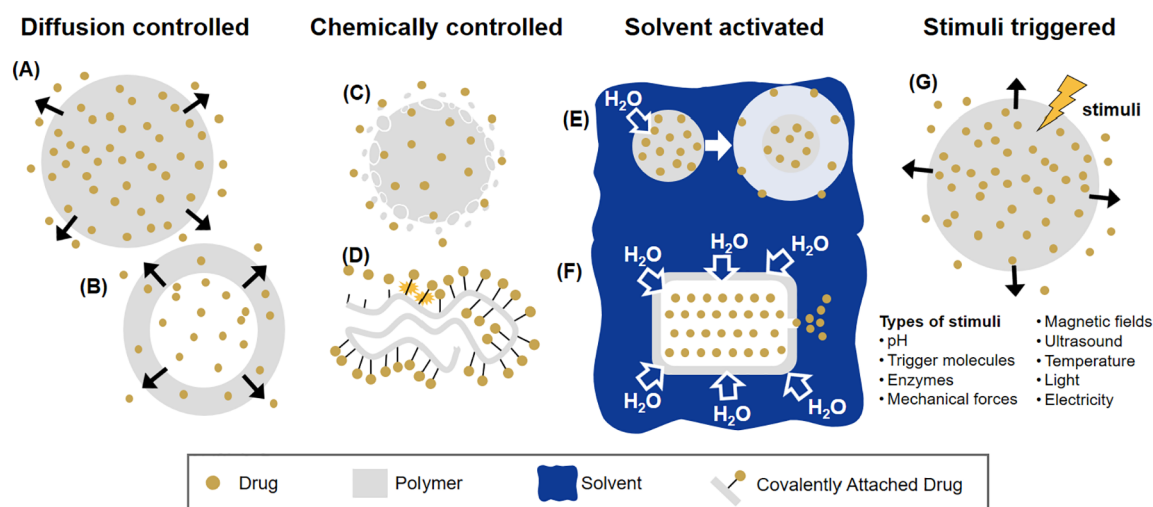


Figure 3. Drug delivery systems that can be used to deliver molecules in a controlled manner. Diffusion controlled systems can be (A) matrix or (B) reservoir based. In chemically controlled systems, the drug can be released by either (C) surface erosion or (D) pendant-chain systems. Solvent-activated systems release the drug by (E) swelling or by (F) osmotic pressure. (G) Stimuli-triggered systems release the drug upon some external or environmental stimuli. Reproduced with permission from ref 77. Copyright 2021 Elsevier.

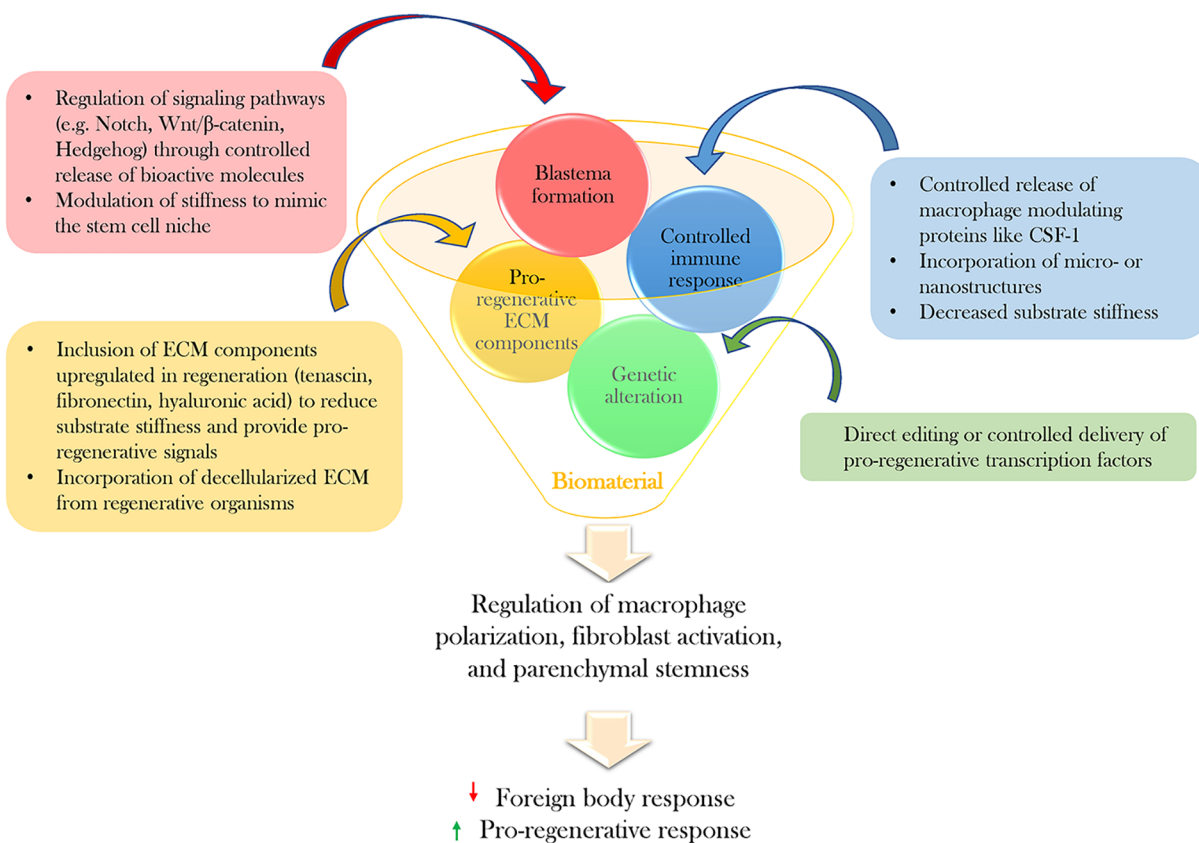


Figure 4. Potential biomaterial design strategies inspired by regenerative organisms to promote pro-regenerative responses to implants. We have identified blastema formation, control of the local immune response, and inclusion of pro-regenerative ECM components as characteristics most crucial to a regenerative response. Genetic alteration of cells near the biomaterial through direct or indirect means may help modify the wound microenvironment and promote these characteristics. Incorporation of components that promote blastema formation, control of the local immune response, and a pro-regenerative ECM into biomaterial implants may improve the longevity of the implant by preventing FBR.

3.3. Controlled Release of Molecules. Our ability to modify biomaterial properties also presents opportunities for the controlled delivery of molecules that can mediate the host cell phenotype. There are several categories of drug delivery

systems that can be applied to biomaterials, which are summarized in Figure 3 and include diffusion controlled, chemically controlled, solvent activated, and stimuli triggered release. Diffusion-controlled release is mediated by a

molecule's diffusion through polymer chains or via channels or interconnected pores, and diffusion of the molecule can be altered by design aspects such as chain rigidity and cross-linking density. Chemically controlled release of drugs or molecules is mediated by degradation of the biomaterial or degradation of pendant chains that connect the molecule to the bulk biomaterial. Chemically controlled release depends on factors like the chemical bond stability, hydrophobicity, porosity, and geometry of the biomaterial. Solvent-activated release of molecules is mediated by swelling or osmosis and depends on penetration of water into the biomaterial. Stimuli-triggered release systems are designed to be reactive to various external factors like pH, temperature, or even the migration of cells (e.g., MMP-degradable cross-links). For more detailed explanations of drug delivery via biomaterial systems, readers are directed to reviews by Silva et al. and Langer and Peppas.^{77,78} These biomaterial release systems allow for local delivery of a drug or molecule, with systemic toxicity or off-target side effects that may arise from systemic delivery to the body.^{79,80}

Biomaterial-based delivery also protects small molecules or proteins from degradation until they are released by serving as a barrier between the molecule and the immune system and is accomplished through functionalization or encapsulation of the molecule or protein by the material.⁷⁸ For example, RNA is prone to degradation due to its chemical instability and the presence of circulating nucleases.⁸¹ Both natural and synthetic biomaterials have been used to deliver RNA to a specific location in the body, preventing its untimely degradation.^{82–84}

4. POTENTIAL STRATEGIES FOR BIOMATERIAL DESIGN INSPIRED BY REGENERATIVE ORGANISMS

In the following sections, we will introduce molecules that may be suitable for mediating FBR, inspired by what has been learned from regenerative organisms. Specifically, we know that blastema formation, controlled immune responses, and pro-regenerative ECM components are features observed in regenerative organisms that enable these organisms to undergo regeneration and avoid fibrosis. Based on these regenerative aspects, we propose certain strategies that can potentially enhance the prospects of an implant for host integration with minimal FBR (Figure 4).

4.1. Reprogramming Resident Cells to Mediate Regeneration. A major obstacle faced in mammalian regeneration is the lack of progenitor cell proliferation and differentiation to remodel damaged tissue. Stem cell therapy is commonly explored in regenerative medicine research as a source of additional multipotent, proliferative cells; however, stem cell delivery to replace functional resident cells has been largely unsuccessful and is associated with complications such as nephrotoxicity,⁸⁵ pulmonary embolism,⁸⁶ thromboinflammation,⁸⁷ and the risk of developing tumorigenesis after stem cell transplantation.⁸⁸ Regenerative organisms inform us that there are other ways to initiate regeneration that are independent of exogenous stem cell populations. Digit tip regeneration in rodents tells us that differences in the availability of endogenous progenitor cells are not a limiting factor to regeneration, but instead that the environment must be appropriate to direct surrounding cells.⁹ Regenerative organisms can reprogram mature cells to repair damaged tissue through dedifferentiation, a process that involves the re-entry of mature cells into the cell cycle, the acquisition of a stem cell-like phenotype, and the expression of stem cell

markers. These events can accomplish this through alterations in the wound environment. For example, cell-free ECM from MRL mouse blastema cells caused improved healing and an increased presence of pigmented hair in a C57BL/6 mouse dorsal wound, demonstrating that regenerative signals in a nonregenerative wound environment can impact healing.⁸⁹ By taking cues from the wound environment of regenerative organisms, we may be able to harness similar healing mechanisms in humans.

Regenerative organisms exhibit epimorphic regeneration, including invertebrates such as planarians, annelids, starfish, and jellyfish;⁹⁰ primitive vertebrates such as zebrafish⁹¹ and axolotls;⁹² and even limited occurrences in mammals such as a digit tip in rodents⁹ and ear pinna in rabbits,^{4–6} MRL mice,^{7,8} and *Acomys*.¹ Blastema formation upon injury is the prominent feature associated with epimorphic regeneration. A blastema is broadly defined as a mass of undifferentiated “blast” cells that migrates and proliferates to heal the damaged tissue. Initially, all blastemal cells were thought to have a pluripotent origin, but they have been found to originate from a range of cell sources in addition to stem cells. In fact, the source of cells in a blastema varies in different organisms, and multiple sources of cells can be responsible for blastema formation at different locations in the same organism. For example, in planaria, pluripotent stem cells (neoblasts) are responsible for the formation of the blastema.⁹³ In contrast, the dedifferentiation of mature cells forms the blastema in zebrafish.^{94,95} In adult newts, limb regeneration is mediated by the dedifferentiation of myofibers,⁹⁶ whereas transdifferentiation of epithelial cells leads to the regeneration of eye lens.^{97,98}

The multiple routes adopted by regenerative organisms in epimorphic regeneration suggest that biomaterials could be leveraged to deliver molecules such as growth factors and reprogram nearby cells to dedifferentiate or transdifferentiate into blastema-like cells and induce regeneration in non-regenerative organisms. This idea has been recently supported by Murugan et al., who triggered limb regeneration in adult *Xenopus laevis* by delivering a cocktail of small molecules that increased the presence of blastemal markers.⁹⁹ Transcriptomic analysis of the tissue buds demonstrated enrichment of the Wnt, Notch, and Hedgehog signaling pathways, which are all linked to regeneration and may be used to induce blastema-like structures.

The Wnt/ β -catenin pathway is involved in most examples of regeneration, including hair follicle regeneration in mammals.¹⁰⁰ The importance of the Wnt/ β -catenin pathway lies in part in its regulation of dedifferentiation, as seen in human epidermal cells,¹⁰¹ articular chondrocytes,¹⁰² and the digit tips of MRL mice.¹⁰³ Using transgenic zebrafish cell lines, Stoick-Cooper et al. showed that Wnt/ β -catenin signaling is required for the formation and subsequent proliferation of blastema in zebrafish tail fin regeneration, and that activating the Wnt/ β -catenin pathway augmented regeneration. This improved regeneration was thought to be caused by FGF signaling, which is an essential regulator of progenitor cell formation in limb and fin regeneration.¹⁰⁴ Inhibition of FGF signaling blocks blastema formation in zebrafish fin regeneration,^{105,106} and introducing FGF-10 into a nonregenerative organism can have pro-regenerative effects.¹⁰⁷ Yokoyama and Tamura introduced FGF-10 into nonregenerative limb stumps of *Xenopus laevis* and found that regeneration was stimulated and there was an increase in *Msx-1* positive undifferentiated blastemal cells, although it was unclear whether these cells

were derived through dedifferentiation of mature cells or stem-like reserve cells.¹⁰⁷ Interestingly, not all Wnt-associated upregulation is conducive to regeneration. While the Wnt/ β -catenin pathway is crucial for the formation and maintenance of the blastema, β -catenin-independent Wnt signaling through Wnt5b represses regeneration, demonstrating the complexity of Wnt pathways.¹⁰⁴

The complexity of Wnt signaling means the pathway is not an easy target. It is highly conserved throughout metazoans and is strictly regulated to maintain homeostasis. It is involved in widespread cellular functions including proliferation, stem cell pluripotency, and cell fate during embryonic development,¹⁰⁸ and its misregulation can cause human pathologies including cancer.¹⁰⁹ Therefore, direct activation of the pathway should be avoided. A potential alternative is downregulating negative feedback loops associated with Wnt/ β -catenin. CXXC-type zinc finger protein 5 (CXXC5) has recently emerged as a negative feedback regulator of the Wnt/ β -catenin pathway that regulates processes such as cutaneous wound healing¹¹⁰ and cellular differentiation.¹¹¹ Disruption of the CXXC5-dishevelled (DVL) interaction with a competitor peptide was shown to stimulate hair regrowth and wound-induced hair follicle neogenesis in mice.¹¹² Furthermore, topical application of a competitor peptide for >6 months did not result in melanoma or other skin pathologies, supporting the safety of targeting negative feedback loops over direct Wnt/ β -catenin activation.¹¹³ The delivery of this peptide or a similar peptide via a biomaterial may further increase the safety of this mechanism through local and controlled release.

Other signaling pathways that could be leveraged to induce blastema-like structures include the Notch¹¹⁴ and Hedgehog pathways.^{115,116} Both coordinate with Wnt signaling during regeneration.^{117,118} The activation of Notch signaling in zebrafish leads to the increased proliferation and expansion of blastema cell markers, while chemical inhibition impairs fin regeneration.¹¹⁹ During myogenesis in mice, Notch signaling is responsible for the dedifferentiation of muscle cells and fibers.¹²⁰ Interestingly, Notch activity needs to be temporally regulated for regeneration. For example, Notch activity is required for the dedifferentiation of Muller glia in chicken retina, but later activation of Notch inhibits differentiation of the newly generated progenitor cells.¹²¹ Similarly, Notch activation is required for the expansion of the satellite cell population in human muscles, but later in myogenesis, the notch pathway is downregulated and newly differentiated muscle cells develop.¹²² Biomaterials pose an option for spatial and temporal control of Notch pathway through controlled release of Notch activating molecules. Inhibition of Hedgehog signaling also impairs regeneration, but for a different reason. Singh et al. treated amputated newt limbs with cyclopamine, an inhibitor of Hedgehog signaling, and did not see any differences in the dedifferentiation stage between the cyclopamine-treated and control limbs when analyzing the tissue with transmission electron microscopy. However, inhibition of Hedgehog signaling did result in halted cell cycle progression and migration of dedifferentiated cells in newt limb regeneration.¹¹⁸

Like the Wnt/ β -catenin pathway, inactivation or undesired activation of these pathways can be detrimental to human tissues and must be approached with caution. Aberrant gain or loss in the Notch pathway is associated with developmental disorders, adult-onset diseases, and cancers.¹²³ Overactivation of the Hedgehog pathway is associated with lung fibrosis¹²⁴

and various cancers in humans.¹²⁵ Therefore, a local and controlled release of Notch and Hedgehog activators or agonists is imperative in their successful application. Luckily, a wide range of techniques exist for obtaining controlled release of molecules in biomaterial systems and new advances are constantly being made.¹²⁶

Heparin-binding EGF-like growth factor (HB-EGF) is known to activate the Notch pathway and promote blastema formation in zebrafish,¹²⁷ and has also been identified as a regulator of tissue repair in humans.¹²⁸ Biomaterial platforms can provide local and controlled delivery of Notch activators like HB-EGF and have already demonstrated success. For example, entrapment of HB-EGF in poly glycolic acid and poly lactic acid-based scaffolds by subcritical CO₂ exposure prolonged the local delivery of HB-EGF, leading to an improved tissue-engineered intestine structure.¹²⁹ Similarly, hydrogels containing sulfated hyaluronan enhanced the efficacy of HB-EGF delivery and improved wound healing in a porcine skin organ culture model, presumably by activating the Notch pathway.¹³⁰

The Hedgehog pathway plays a critical role in mammalian bone regeneration as well as cell–biomaterial interactions. For example, micro/nanotextured topographies appear to promote osteogenesis via activation of Hedgehog-Gli1 signaling.¹³¹ Similarly, apatite-coated 3D scaffolds immobilized with Hedgehog-activating-agonist-based nanoparticulate deliver a more favorable environment than the commonly used BMP2-incorporated collagen sponges for osteogenic differentiation *in vitro* and calvarial bone healing *in vivo*.¹³²

These novel biomaterials open the door for modifications of existing materials to deliver regeneration-promoting bioactive components like growth factors, which can modulate the Wnt/catenin, Notch, and Hedgehog pathways and help reprogram host cells to minimize FBR. Another option for reprogramming host cells is functionalizing biomaterials with transcription factors (or their inhibitors) to aid in modifying the microenvironment and instructing the responses from the native cells.¹³³ Additional strategies for the delivery of genetic modulators like transcription factors via biomaterials are summarized in section 4.4.

4.2. Controlled Immune Response. Understanding the regulation of immune cells in regenerative organisms could also guide us to new targets that aid in reducing FBR. For example, combinatorial strategies using biomaterials and molecular biology techniques such as recombinant protein delivery could mitigate FBR by modulating immune regulators seen in regenerative organisms. Here, we describe the significance of macrophages in regeneration and present strategies inspired by regenerative organisms for minimizing the immune response to biomaterials.

Macrophages are crucial to regeneration as they regulate blastema formation, even in mammalian regeneration.^{41,134} Neonatal mice treated with clodronate liposomes to deplete macrophages were unable to regenerate following myocardial infarction and instead formed fibrotic scars.¹³⁵ Similarly, depletion of macrophages leads to missing or inadequate limb regeneration in axolotls,¹³⁶ and zebrafish with depleted macrophages show aberrant tail fin regeneration,¹³⁷ further indicating the critical role of these immune cells in regulating the regenerative potential of vertebrates. Simkin et al. demonstrated that clodronate liposome inhibited regeneration in *Acomys* ear pinna, but re-epithelialization and blastema formation occurred once treatment was stopped.¹³⁸ This

rescue of regeneration was also demonstrated in axolotl once clodronate liposome treatment was stopped.¹³⁶ In macrophage-depleted axolotls, the production of ECM-degrading enzymes was reduced and there was increased deposition of collagen I, collagen IV, and α -smooth muscle actin, eventually leading to the formation of scar tissue.¹³⁶ These findings demonstrate that macrophages are instrumental to re-epithelialization, blastema formation, and ECM composition during regeneration.

Although complete ablation of macrophages has a negative impact on regeneration, there is evidence that the macrophage population and phenotype are heavily controlled in regeneration. Healing embryonic wounds are associated with lower numbers of differentiated inflammatory cells, while adult wounds demonstrate increased numbers of monocytes and macrophages that persist for longer.¹³⁹ Classically activated M1 macrophages were found to be present in *Mus* granulation tissue following an ear pinna punch but were located proximal to the wound site and did not infiltrate the blastema.¹³⁸ Interestingly, Godwin et al. found unusually early activation of anti-inflammatory macrophage signaling in regenerating axolotl limbs that occurred simultaneously with pro-inflammatory signals.¹³⁶ This is different than the M1 to M2 phenotype switch that generally occurs at later stages in mammalian wound healing. In addition, all cytokines and chemokines had returned to baseline levels by the late blastema stage.¹³⁶ This transient immune response intuitively makes sense, because the prolonged presence of either pro- or anti-inflammatory macrophages is associated with aberrant wound healing.^{19,140–142}

Biomaterials may be used to direct a transient immune cell response in humans to try to promote a regeneration permissive environment. For example, taking cues from axolotl limb regeneration, the early delivery of IL-4 and IL-13¹³⁶ following biomaterial implantation may result in an earlier infiltration of anti-inflammatory macrophages in concert with pro-inflammatory macrophages recruited by tissue damage and better mimic a regenerative environment. Hachim et al. demonstrated this concept by developing a nanometer-thickness chitosan and dermatan sulfate bilayer coating that was used on a polypropylene mesh and released IL-4 at the early stages of the host response. The release of IL-4 could be tuned by changing the number of layers of coating utilized.¹⁴³ When the coated mesh was implanted subcutaneously in a mouse, increased recruitment of macrophages, diminished fibrotic capsule thickness, and decreased collagen fiber thickness were observed.¹⁴³

Another potential target for creating a regeneration permissive environment is colony stimulating factor 1 (CSF-1), a secreted cytokine that plays a role in the proliferation and differentiation of immune cells during regeneration. Mutant CSF-1 zebrafish models display impaired angiogenesis,¹⁴⁴ an abnormality known to inhibit zebrafish regeneration^{145,146} and mammalian wound healing.¹⁴⁷ CSF is also needed for liver regeneration in mammals,^{148,149} and short-term treatment using a recombinant CSF-Fc conjugate is being explored as a therapeutic treatment to enhance liver regeneration.¹⁵⁰ Although important for regeneration, the contribution of CSF-1 to the central inflammation pathway can also cause subsequent fibrosis and has been identified as a key component of FBR in rodents and nonhuman primates.¹⁶ Doloff et al. demonstrated that inhibition of the CSF-1 receptor with a small molecule prevented the fibrotic response to implanted

alginate, glass, and polystyrene microspheres in mice without eliminating other functions of macrophages.¹⁶ Building on these findings, the functionalization of biomaterials with CSF-1 agonists could be a potential strategy to promote implant integration. Once again, our knowledge of regenerative organisms tells us that fine control, not complete inhibition, should be the goal, as CSF-1 plays a role in proper wound healing. The best route to ensure functional incorporation of the implanted biomaterial may be a delayed release of CSF-1 or other related factors to promote normal macrophage activity in the early stages of wound healing but inhibit persistent inflammation.

4.3. Pro-regenerative ECM Components. ECM-based biomaterials are commonly used to repair or remodel injured tissues because they provide cues that can orchestrate the host cell response. These cues can include appropriate mechanical properties, the release of bioactive molecules, and the promotion of cell attachment and migration. Examples of ECM-based materials and fabrication strategies have been reviewed by Hussey et al.¹⁵¹ The functions of ECM-based biomaterials have also been applied to synthetic implants by adding an ECM coating to improve outcomes. For example, coating titanium implants with ECM components enhances osteointegration and improves implant stability.^{152,153} Similarly, a collagen or laminin coating was shown to support epithelialization of synthetic corneal implants, which is important for maintaining a healthy cornea after implantation.¹⁵⁴ However, despite years of research and application, biomaterials composed of mammalian ECM have not unequivocally demonstrated modulation of FBR.²⁷ We propose that the ECM of regenerative organisms may provide unique structural and chemical cues to modulate FBR, as these organisms respond to tissue insults with functional repair instead of fibrosis and persistent inflammation.

There is evidence that the ECM composition of regenerative organisms differs from that of nonregenerative counterparts, suggesting there may be pro-regenerative matrix compositions that direct better tissue reconstruction following wounding. The importance of matrix composition to regeneration was demonstrated by Vorotnikova et al., who showed that cell-free ECM from the blastema or an MRL mouse caused improved healing and an increase in pigmented hair in B6 mouse dorsal wounds. The MRL blastema cells expressed high levels of tenascin C, as well as Thy1 and thrombospondin.⁸⁹ This finding demonstrates the right ECM composition can improve regeneration, while Satoh et al. have demonstrated that the wrong ECM composition can inhibit regeneration. The application of a decellularized collagen patch in an accessory limb model in axolotls prevented blastema formation and limb regeneration.¹⁵⁵ During epimorphic regeneration of ear hole punches, *Acomys* regenerative ECM is dominated by fibronectin and tenascin, while scarring *Mus* ECM is dominated by an early and large upregulation of collagens col1a1 and col3a1 .¹⁵⁶ Fibronectin and tenascin are also upregulated in the blastema of regenerating axolotl and newt limbs and have been shown to promote cell migration and proliferation.^{157–159} A similar fibronectin- and tenascin-rich provisional matrix has been identified following zebrafish fin amputation¹⁶⁰ and myocardial infarction in zebrafish and newts.¹⁶¹ Tenascin C was reported to significantly increase newt cardiomyocyte cell cycle re-entry in vitro following cardiac injury.¹⁶¹ Both fibronectin and tenascin are ECM glycoproteins, and their interaction has been shown to alter the

expression of genes like collagenase in cultured fibroblasts, suggesting an additional role in tissue remodeling.¹⁶² In addition to fibronectin and tenascin, hyaluronic acid (HA) contributes to the transitional extracellular matrix formed during axolotl limb regeneration.¹⁶³ Axolotl retina also contain a higher HA content compared to mammals which are unable to regenerate the retina,¹⁶⁴ and HA synthesis is required for zebrafish tail fin¹⁶⁵ and heart¹⁶⁶ regeneration.

Each of these three ECM components—fibronectin, tenascin, and hyaluronic acid—have been incorporated into biomaterials in the context of wound healing. Fibronectin and the short peptide RGD derived from fibronectin are commonly used coatings to promote cell adhesion, migration, and proliferation. In a murine wound healing model, fibronectin nanofiber scaffolds were shown to promote recovery of native tissue architecture, including *de novo* hair follicle regeneration.¹⁶⁷ Supplementation of decellularized lung tissue with tenascin C, in combination with fibrillin-2, was shown to increase basal epithelial stem cell migration and tissue injury repair during *ex vivo* recellularization.¹⁶⁸ Hyaluronic acid is widely used in biomaterials due to its versatility and role in wound healing, and new advances are constantly made in fabrication techniques. Porous nanofiber HA scaffolds were recently shown to promote faster wound closure and tissue repair in full-thickness skin wounds in mice compared to denser HA scaffolds.¹⁶⁹ Other examples of recent advances in HA biomaterial systems for wound healing were reviewed by Yang et al.¹⁷⁰ These successful applications of pro-regenerative ECM components in the context of wound healing show promise for their future use in the functional integration of implants.

ECM components are now being isolated directly from regenerative organisms, opening the door for their direct use in biomaterials. For example, marine collagen has gained popularity as a replacement for mammalian-derived collagen to avoid transmission of diseases such as bovine spongiform encephalopathy. Collagen isolated from jellyfish has been shown to mediate fibrosis following implantation *in vivo* compared to bovine collagen.¹⁷¹ In addition to utilizing individual ECM components, decellularized extracellular matrices (dECMs) are a popular approach in tissue engineering due to their complex composition and tissue-specific structure. Certain protocols support retention of matricellular proteins like tenascins¹⁷² and glycoproteins like hyaluronic acid¹⁷³ during decellularization, suggesting that dECM from regenerative organisms may maintain pro-regenerative cues. This has been confirmed by Chen et al, who showed that dECMs derived from zebrafish hearts were shown to induce the proliferation of cardiac precursor cells in adult mice postacute myocardial infarction.¹⁷⁴ It stands to reason that dECMs from other regenerative organisms may also support regeneration when transplanted into mammals. Axolotl skin¹⁷⁵ and retinal tissue¹⁶⁴ have been decellularized and characterized, and a method to isolate ECM from planaria has recently been developed.¹⁷⁶ As new decellularization protocols for regenerative organisms are designed, our understanding of pro-regenerative ECM compositions and their impact on cell behavior will increase, paving the way to their implementation in biomaterial designs or coatings to promote functional incorporation of the material.

4.4. Genetic Modulation to Regulate Regenerative Signaling Pathways. Thus far, we have provided examples of bioactive molecules that could be used to modulate pathways

associated with regulating blastema formation, immune response, and ECM maintenance in regenerative organisms. There are also interesting gene-level targets that could be locally modulated by regulating their expression directly or by regulating relevant transcription factors.

Two regeneration-specific proteins expressed in salamanders are *Prod1* and newt-anterior gradient protein (nAG), which are crucial for blastema formation and proliferation of progenitor cells.^{177,178} *Prod1* and nAG and genes suitable for humans have been designed, cloned, and successfully expressed in a distal phalanx model in mice, resulting in a quicker and more mature bony regeneration.¹⁷⁹ The same group previously demonstrated that injection of recombinant nAG protein reduced scar hypertrophy in rabbits¹⁷⁹ and liver fibrosis in rats,¹⁸⁰ supporting the relationship between regenerative mechanisms and antifibrotic mechanisms. Releasing these recombinant proteins in a controlled manner from biomaterial implants may be able to induce a more stable, sustained, and localized regenerative response in mammals.

While nAG and *Prod1* are pro-regenerative genes, certain genes are reported to be detrimental to regeneration. The MRL mouse was created in the laboratory using several outbred and inbred experiments, which resulted in the loss of the p21 encoding gene (*Cdkn1a*) and a mutation in the *Exo1* gene. Interestingly, these unique features granted them an improved ability to heal ear punch wounds,¹⁸¹ articular cartilage damage,^{182,183} and amputated digits.¹⁸⁴ Creating a similar mutation in *Exo1* and downregulating p21 in non-regenerative *Mus* increases their lifespan as well and imparts limited ability to regenerate,¹⁸⁵ giving us hope that it may be possible to induce mammalian regeneration by modulating the specific targets informed by regenerative organisms. The advancement of biotherapeutic micro- or nanocarriers to deliver specific nucleic acids and development of biomaterial-based delivery of CRISPR/Cas-9 widens the horizon of genetic alteration^{186,187} and opens the door for local delivery or targeting of genes that can aid in mammalian tissue repair.

Alternatively, transcription factors (TFs) could be delivered to upregulate pro-regenerative genes like *Prod1* and nAG or downregulate antiregenerative genes. TFs regulate the transcription of DNA to RNA and can regulate specific genes by binding to nearby DNA. Introduction of transcription factors is a commonly used method for reprogramming somatic cells into pluripotent cells *in vitro*.^{188–190} Delivering tissue-specific transcription factors can also reprogram mammalian fibroblasts to transdifferentiate into cardiomyocytes,¹⁹¹ endothelial cells,¹⁹² and neurons¹⁹³ without first becoming progenitor cells. Transcription factor-mediated reprogramming has also been achieved *in vivo*. For example, Zhou et al. demonstrated that viral delivery of three transcription factors could reprogram adult pancreatic exocrine cells into β -cells in adult mice. The resulting β -cells were able to ameliorate hyperglycemia by secreting insulin and expressed genes essential for β -cell function.¹⁹⁴ Similarly, the induction of SOX2 resulted in the conversion of astrocytes to neuroblasts in the brains of adult mice.¹⁹⁵ These neuroblasts were then able to differentiate into functionally mature neurons.¹⁹⁵ Resident cell types can become more plastic following injury,^{196,197} suggesting they are more amenable to fate switch in a postimplant microenvironment. Transcription factors can be delivered indirectly through RNA or DNA or directly as part of a macromolecule containing cell-penetrating peptide.¹⁹⁸ Advancement in mRNA delivery in humans has increased exponentially as a

result of coronavirus-related mRNA vaccine research,¹⁹⁹ and these delivery strategies likely could also be harnessed to mitigate FBR. Biomaterial-based gene therapies can overcome issues with stability and delivery by protecting encapsulated nucleic acids and providing local release that can be controlled by altering the method of drug loading and the degradation of the biomaterial.^{82–84,200,201}

One potential TF candidate for delivery via biomaterial implants is the *Twist* protein family, regulated by the *Twist* gene, which plays an important role in embryonic development. Its orthologue in axolotls, *AmTwist*, is involved in determining the lineage and differentiation of cells. *AmTwist* is upregulated during regeneration but downregulated during the initial step of blastema formation in axolotls and its expression is regulated by the neural signaling and bone morphogenetic protein (BMP-2).²⁰² *Twist3*, an ortholog of mammalian *Twist* genes, is involved in muscle regeneration in zebrafish,²⁰³ reflecting the relevance of twist factors in regulating and facilitating regeneration. The *Twist* family regulates self-renewal and lineage determination in humans as well. Inducing higher expression of *Twist* in bone marrow-derived human mesenchymal stem cells enhances the proliferation of stem cells, but overexpression decreases their ability to differentiate, indicating a need for tight regulation to obtain the desired cell behavior.²⁰⁴ This need for tight regulation is emphasized because overexpression of *Twist* in humans is also associated with cancer metastasis.²⁰⁵ Biomaterials could be ideal for temporal regulation and local delivery of *Twist* proteins and other TFs, as they can release molecules in a dose- and time-controlled manner.

Although transcription factors have been used successfully to reprogram cells both *in vitro* and *in vivo*, it is important to note that the delivery of tissue-specific transcription factors may not be sufficient to promote dedifferentiation. Epigenetic modifications including DNA methylation, histone modifications, and chromatin structure exist to maintain somatic cell identity following development and lineage specification.¹⁹⁷ These epigenetic modifications can interfere with reprogramming, and not all transcription factors are capable of modifying epigenetics.¹⁹⁰ DNA methylation can be preventative for reprogramming, as hypomethylation is required for activation of pluripotency.²⁰⁶ Therefore, the use of DNA methyltransferase inhibitors can be used in combination with transcription factors to improve reprogramming efficiency.²⁰⁷ In addition to maintaining cell lineage specification following differentiation, epigenetic modifications like hypermethylation can stably modify the activation of myofibroblast and contribute to the pathogenesis of fibrosis.¹⁴² For example, hypermethylation of the *RASAL1* gene was associated with myofibroblast presence in fibrotic kidneys of mice.²⁰⁸ Knocking out the methyltransferase *Dnmt1* in mice was able to ameliorate kidney fibrosis.²⁰⁸ Similarly, hypermethylation in the *Thy-1* promoter of fibroblastic cells was found in the IPF lung of mice and could be modified with DNA methyltransferase inhibitors to protect mice from pulmonary fibrosis.²⁰⁹

Designing biomaterials functionalized to modulate genes important to fibrosis and regeneration may offer exciting ways to direct the response from the host and promote functional integration in lieu of fibrosis.

5. SUMMARY AND FUTURE DIRECTIONS

The way we approach biomaterial design is constantly changing, and it may continue to change due to our growing

understanding of regenerative organisms. For example, biomaterials were traditionally designed to be inert or anti-inflammatory, with the development of steroid-releasing polymers beginning in the 1960s.^{210,211} However, while anti-inflammatory molecules like steroids can downregulate inflammation, disruption of normal healing and reconstruction can also occur.²¹² Our knowledge of regenerative organisms informs us that an initial inflammatory response is crucial for successful regeneration and blocking this response can have negative effects.^{135,136,138} Therefore, biomaterials would be better served to *direct* the host immune response, not *inhibit* it, as with recent immunomodulatory materials.²¹³ While a delayed phenotype shift from M1 to M2 has been a recent focus of biomaterial design,^{28,42,214} design criteria informed by regenerative organisms may warrant coordinated recruitment of both pro- and anti-inflammatory macrophage phenotypes.

A similar paradigm shift may be necessary in the future regarding the role of reactive oxygen species (ROS) in FBR. Many biomaterials are currently designed to scavenge ROS and downregulate oxidative stress due to their roles in inflammation.²¹⁵ However, ROS production is crucial for successful regeneration. ROS is significantly elevated in the blastema of *Acomys*,¹³⁸ tadpoles,²¹⁶ and zebrafish, and sustained production of ROS is essential for blastema formation in zebrafish following fin amputation.²¹⁷ Reducing ROS levels following tadpole tail amputation reduced cell proliferation, impaired regeneration, and decreased Wnt/ β -catenin signaling.²¹⁶ Similarly, inhibition of ROS biosynthesis by neural stem cells in the adult newt brain counteracts proliferation of neural stem cells and neurogenesis.²¹⁸ These results suggest that ROS scavenging biomaterials may be inhibiting proper healing and tissue reconstruction following biomaterial implantation, similar to the delivery of steroids. However, it is also important to note that regenerative organisms seem better equipped to detoxify ROS and are less susceptible to ROS-induced senescence.²¹⁹ A closer look at stress-sensing mechanisms in regenerative and nonregenerative organisms is needed to better understand the proper path for ROS-modulating materials.

In summary, taking cues from regenerative organisms may enable researchers to push the host response toward functional incorporation of biomaterials and away from fibrotic encapsulation. This hypothesis arises from the similarities between a fibrotic response to tissue injury and the processes that lead to the fibrotic encapsulation of biomaterials. Our assessment of FBR in *Acomys* is a crucial first step and supports the notion that fibrosis and FBR are intimately linked.¹⁴ Regenerative organisms respond to tissue injury through changes to the wound environment that promote blastema formation, immune cell modulation, and pro-regenerative ECM. The field's ability to tune the physical and chemical properties of biomaterials presents opportunities to alter the postimplantation environment and direct pro-regenerative functions and pathways. Furthermore, the ability for biomaterials to deliver molecules in a local and controlled manner provides additional control over the implant microenvironment. Strategies derived from regenerative animals could be leveraged for future biomaterials to reprogram the microenvironment of nonregenerative mammals and awaken lost regenerative abilities, improving the outcomes of implanted materials.

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Notes

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ABBREVIATIONS

BMP2, bone morphogenetic protein 2; Cdkn1a, cyclin dependent kinase inhibitor 1A; CSF-1, macrophage colony stimulating factor 1; CXXCS, CXXC-type zinc finger protein 5; dECM, decellularized extracellular matrix; ECM, extracellular matrix; FBGCs, foreign body giant cells; FBR, foreign body response; HA, hyaluronic acid; HB-EGF, heparin-binding EGF-like growth factor; IFN- γ , interferon gamma; IL, interleukin (e.g., IL-1 β); MMP3, matrix metalloproteinase 3; MMP9, matrix metalloproteinase 9; NF- κ B, nuclear factor kappa B; RGD, arginylglycylaspartic acid; TF, transcription factor; TGF- β , transforming growth factor beta; TH17, T helper 17; TNF- α , tumor necrosis factor alpha; TRVP4, transient receptor potential cation channel subfamily V member 4

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